Evaluation & Testing

Scope | the Eval action items from the last two meetings, and where each one is answered

Action item	What we did	Slide
Run round 1 in week 8 (minutes #5)	RAG ran our scoring on Gold Set v2 (11/9); we fixed the anchors it flagged	2
Run one question set on RAG and GraphRAG, side by side	All 84 Microsoft 10-Q questions through each team's own code, same filing, top 8	3
Show where GraphRAG does better, and where RAG does	Plus the 9 Exhibit 21 relationship questions (traversal)	4
Label the routing chart's axes; define the keyword router	Chart redrawn by expected route, axes titled	5
Explain how unneeded calls per question are measured	Definition with a worked example	5
Concrete routing examples for the demo	Seven questions, three routers, calls counted	6
Say accuracy is on the current 155 questions only	Stated on every routing and wording number	5-7
Typos and messy wording, for real-user accuracy	427 rewrites of the 155 questions through gate, router and both retrievers	7

All figures use the current 155 Gold Set v2 questions

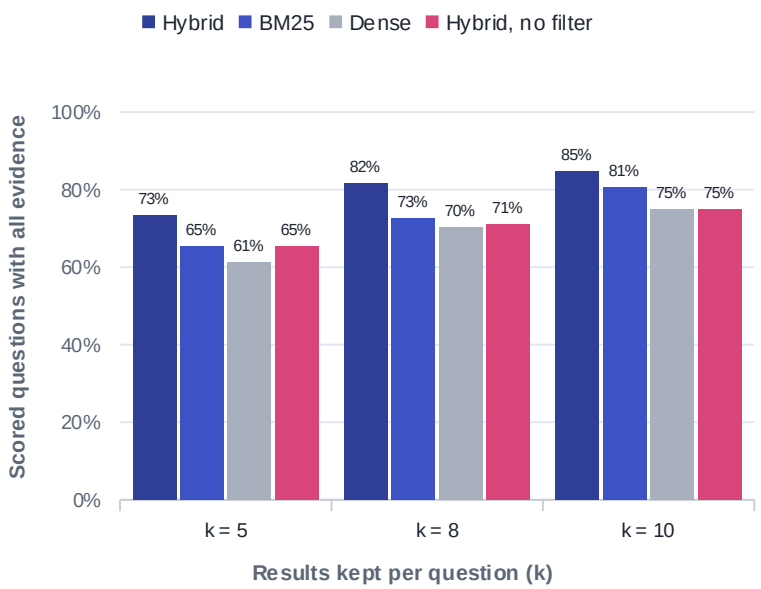
104 finance data, 31 text, 11 relationship, 8 no tool, 1 both. Microsoft has 92 of them, so per-route numbers on 1-11 questions are examples, not estimates. They will move as questions are added.

Pinned: RAG main @ 5bfafeb (11/5), GraphRAG main @ 4e89acb (11/11) and Exhibit 21 branch @ c21fe89, Data @ f8d18e2. No LLM is used in any result; everything except RAG's round-1 run (slide 2) re-runs here exactly.

Source: Week 8/README.md; scripts/run_all.sh (83 tests pass)

Evaluation & Testing

Round 1 | retrieval on Gold Set v2, run by RAG with our scoring rules (11/9): 105 of 124 found



83/93 number questions; prose 22/31 (hybrid, k = 10, hit must come from the right filing)

34/37 single-company questions on the nine other companies, vs Microsoft 71/84: adding nine companies did not hurt retrieval

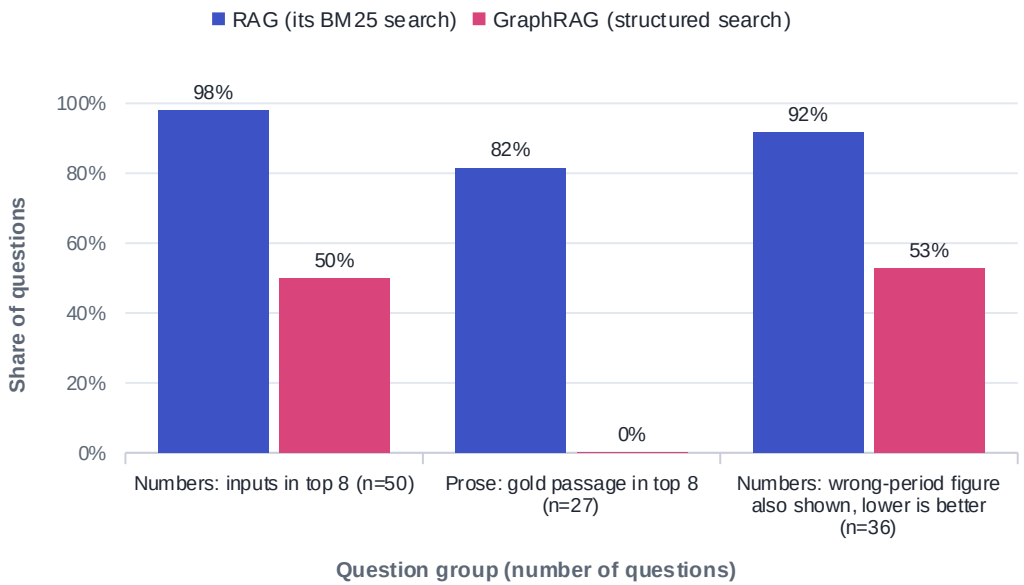
Why 19 were missed	n	Who fixes it
Cross-company ranking: 10-20 figures	3	Agents: one search per company
Our anchors were wrong (Gold v0 conversion)	4	**Eval: fixed in v2.1 (all 22 converted anchors)**
Needs the 10-K and the 10-Q together	3	Agents: one search per form
Fusion pushed it just past k = 10	6	RAG: candidate k, fusion weights
All three methods missed	3	RAG: ranking; 1 is truly multi-part

Also found: a wrong-period figure sits in the top 8 for 90.5% of the 42 number questions that have one, so the reader must be told which column to take; fusion scores look the same for hits and misses, so 'weak evidence' has to come from structure (empty result, no filter match, BM25 and dense disagree).

Source: RAG team, round-1 results 11/9 (KPMG_RAG scripts/run_round1_eval.py), transcribed in Week 8/data/round1_rag_2026-11-09.json; Gold Set v2.1 in Week 8/GOLD_SET_V2.md. Evidence level; no reader yet

Evaluation & Testing

RAG vs GraphRAG | the same 84 Microsoft 10-Q questions, each team's own code on the same filing



Head to head on the 77 answerable questions

	GraphRAG has it	GraphRAG misses
RAG has it	25	46
RAG misses	0	6

How it was run. RAG: its chunker, index and Retriever.search with the question's filters. GraphRAG (main, 11/11, via graphrag.backends): its fact loader built the store from Data's copy of the filing (1,317 facts), then structured_search. Its graph arm (696 claims for this filing) runs on an Aura database our workspace cannot reach, so GraphRAG will run our script for it.

Neither side returned an empty result on the 7 unanswerable questions, so abstaining stays with the Agents' evidence check.

Read: RAG's search finds the needed numbers and passages more often; GraphRAG's facts carry their exact period, so wrong-period figures show up less. Evidence level only: no reader model yet.

Source: Week 8/results/side_by_side.json (scripts/run_side_by_side.py). RAG's hybrid with its offline stand-in for the dense model: numbers 82%, prose 85%

Evaluation & Testing

Where each does better | GraphRAG on relationships across filings, RAG on reading the filing itself

9 relationship questions (Exhibit 21)

9/9 answered exactly by traversal of GraphRAG's own Exhibit 21 rows (parent HAS_SUBSIDIARY subsidiary): counts, joins, ownership checks

0/9 reachable by RAG today: it indexes each filing's main report, and Exhibit 21 is a separate document

8/9 with Exhibit 21 added as text, RAG's search brings every needed row; the counting and joining is then left to the reader (Apple and Micron's shared countries: 15 of 31 rows)

Cross-check: GraphRAG's Exhibit 21 pipeline (11/8) and our gold reads agree on 62 of 62 rows, names and jurisdictions. Rows are GraphRAG's; the traversal ran in memory here, since their Neo4j is not reachable from our side.

	Better at	Evidence (Microsoft 10-Q, top 8)
GraphRAG	Relationship questions	Counts and joins come out exact (left)
	Exact periods on numbers	Wrong-period figure shown 53% vs RAG 92%
RAG	Prose and explanations	22/27 vs 0/27: structured search holds numbers only
	Finding the right number	49/50 vs 25/50

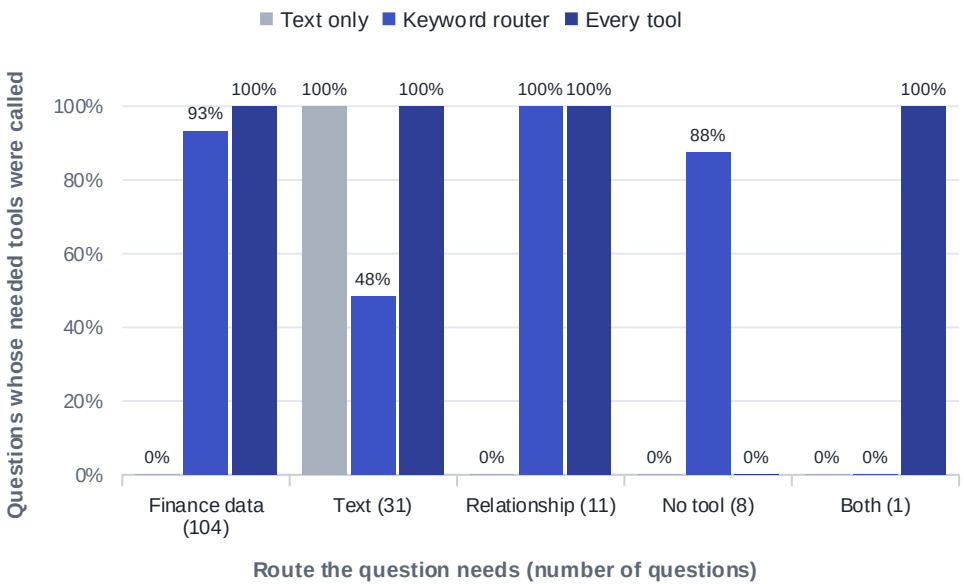
Where GraphRAG's 25 number misses come from (each has a fix)
13 ranked below 8: in 6, 'three months' or 'six months' matches Microsoft 365 ('ThreeSixFive') rows, so total revenue ranks 10th
10 calculations: one call cannot fetch every input; one call per input would
2 wording: 'current unearned revenue', 'repurchasing common stock' miss their concepts
Rank 1: right number 14/50, a wrong-period number 6/36; GraphRAG flagged this itself

Together: route relationship questions to GraphRAG and narrative ones to RAG; numbers can use either once GraphRAG's ranking fix lands, with GraphRAG's period labels as the check.

Source: Week 8/results/side_by_side.json (part_b, number_miss_diagnosis); GraphRAG ex21_pipeline/output/graph_edges.csv @ c21fe89; data/sec_reads/exhibit21_2026-11-05.json; RAG rag/ingest.py

Evaluation & Testing

Routing | which tools each router calls, by the route a question needs (155 questions)



Keyword router: Eval's stand-in for the Agents' orchestrator, rules only, no LLM. Scope check first (blocked: no tool). Then a relationship word (subsidiary, Exhibit 21, acquire, own) goes to GraphRAG; a number word (how much, revenue, margin, total) to finance data; an explanation word (why, explain, describe, risk) to RAG; both kinds to both.

Unneeded call: a tool called that the question's route does not need. Per question = all unneeded calls / 155. Example: a finance-data question sent to every tool makes 3 calls, 2 unneeded. It is not "text search alone would have done".

Router	Needed tools called	Calls / question	Unneeded / question
Text only	20%	1.00	0.79
Keyword router	84%	1.05	0.25
Every tool	95%	3.00	2.05

On the current 155 questions only: the 'Both' and 'No tool' bars rest on 1 and 8 questions. The keyword router's weak spot is text questions that mention a number word (next slide).

Source: Week 8/results/routing_baselines.json; kpmg_eval/metrics/routing.py; ROUTING_EXAMPLES.md. Routes are Eval's labels, for Agents to review

Evaluation & Testing

Routing walkthrough | seven questions for the demo: what each router calls, and why

Question (shortened)	Needs	Keyword router calls	Because	Unneeded: text only / keyword / every tool
Microsoft's total revenue, three months ended Mar 31, 2026	Finance data	**Finance data**	number word 'total'	1 / 0 / 2
Name Microsoft's reportable segments	Text	**RAG**	no number word	0 / 0 / 2
How many subsidiaries does Micron list in Exhibit 21; how many abroad?	Relationship	**GraphRAG**	relationship word 'Exhibit 21'	1 / 0 / 2
How did Microsoft's OpenAI gains change, and what event drove it?	Both	**RAG (misses)**	no number word	0 / 0 / 1
Microsoft's share price at the close on Mar 31, 2026	No tool	**nothing (blocked)**	scope check: abstain	1 / 0 / 3
What does revenue allocated to remaining performance obligations include?	Text	**Finance data (misses)**	number word 'revenue'	0 / 1 / 2
Calculate operating cash flow less capex ... (use that definition)	Finance data	**Finance data, RAG**	number word 'calculate' and explanation word 'definition'	1 / 1 / 2

For the demo: the first five show each route; the last two show where rules break, which is what the Agents' orchestrator should do better. A miss means a needed tool was not called.

Source: Week 8/ROUTING_EXAMPLES.md, results/routing_examples.json (scripts/routing_examples.py); full questions in data/benchmark_v2_questions.md

Evaluation & Testing

Typos and messy wording | 427 rewrites of the 155 questions, same meaning, scored like the clean ones

Rewrite (n)	Example	Scope check, no company picked	Same, company picked in the UI	Router: needed tools	RAG search	GraphRAG search
Company misspelled (129)	Microsfot, Mircon, Net App	**98% → 19%**	98% → 93%	86% → 81%	93% → 93%	37% → 37%
Finance word misspelled (143)	revnue, opertaing, subsidaries	82% → 82%	99% → 90%	85% → 78%	91% → 80%	30% → 17%
Casual: ticker, lower case (155)	what was msft total revenue for ...	**82% → 9%**	97% → 90%	84% → 77%	92% → 91%	32% → 32%

94%

right decisions on misspelled names once the scope check also matches tickers and near-miss spellings (79% on casual); no change on the clean questions

What drops, and why

Company names: the check knows exact names only, so 'Microsfot' or 'msft' turns an answer into 'which company?'.

Finance words: RAG's search loses 8 and GraphRAG's 9 of 70 questions; keyword matching does not forgive 'revnue'.

Searches ignore name typos because the company comes in as a filter.

For Agents: match tickers and near-miss names before routing, and correct spelling before search. Scores are clean → messy, on the current 155 questions; searches on the Microsoft 10-Q questions only.

Source: Week 8/results/messy_wording.json, data/messy_wording_v1.jsonl (scripts/run_messy_wording.py). Scope check and router are Eval's reference rules, not the orchestrator

Evaluation & Testing

Next steps | after round 1: one ask per team

Team	Ask (from this week's results)
RAG	Share the round-1 chunks privately so we can anchor the last 4 questions and run the side by side on all ten; try candidate k and fusion weights on the 6 near misses; reader API key
GraphRAG	Run our round-1 script with your Aura graph and send the results; share the 20-filing store; accept accession filters in place of fiscal labels; read periods when ranking
Agents	Split cross-company and 10-K + 10-Q questions into one search each; match tickers and near-miss names; run your orchestrator through our routing scorer
Data	Keep the 11/03 freeze (matches our 21/21) through round 1; then load Microsoft's FY2024 and FY2025 10-Ks, already in your inventory (joint ask with GraphRAG)
Eval	Anchors for the last 4 questions once chunks land; check RAG's per-question results against v2.1; answer-level round 1 when a reader can run; grow the 1 'both' and 8 'no tool' routes

Thanks: RAG ran round 1 and traced every miss, which is how we found our anchor errors. RAG's retriever and GraphRAG's fact loader ran here unchanged, and GraphRAG's Exhibit 21 output fed the relationship check.

Source: Week 8/FOR_TEAMS_W8.md (one section per team, with file paths)